mbsolver

Three-Level (Λ) Model

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https://github.com/vahid-anari/mbsolver

This document derives the dimensionless Maxwell–Bloch equations of a Λ-type three-level system solved by `mbsolver.three_level.solve_1d`: two dipole-allowed transitions $3 \to 1$ and $3 \to 2$ share the upper level. It starts from the three-level Hamiltonian and proceeds through the rotating-wave and slowly-varying-envelope approximations to the final dimensionless system, its initial conditions and the scales that connect it to physical values. The model and its application to the 6.7 GHz methanol flare in S255IR-NIRS3 are described in a paper by the authors that is currently under review.

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1 Level Scheme and Hamiltonian

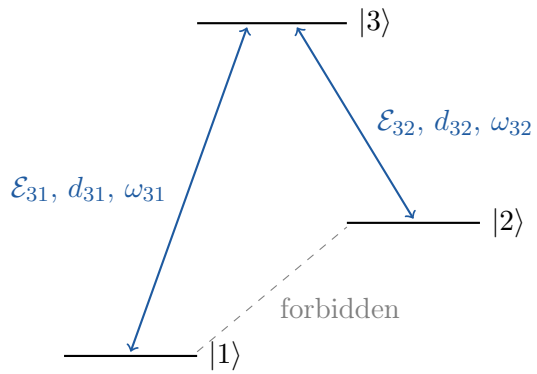


Figure 1: The Λ-type three-level system. The upper level $|3\rangle$ is shared by the dipole-allowed transitions $3 \to 1$ and $3 \to 2$; the transition $1 \leftrightarrow 2$ is dipole forbidden. Level spacings are schematic.

The molecule has three states $|1\rangle$, $|2\rangle$, $|3\rangle$ (Fig. 1). Taking the energy of $|1\rangle$ as zero, the free Hamiltonian is

$$\hat{H}_0 = \hbar\omega_2 \, |2\rangle\langle 2| + \hbar\omega_3 \, |3\rangle\langle 3|, \tag{1}$$

so that the transition frequencies are $\omega_{31} = \omega_3$ and $\omega_{32} = \omega_3 - \omega_2$. With the dipole matrix elements $d_{ij} = \langle i|\hat{d}|j\rangle$, $d_{ij} = d_{ji}^*$, and $d_{12} = 0$ for the forbidden transition, the dipole operator along the field polarization is

$$\hat{d} = d_{13} \, |1\rangle\langle 3| + d_{23} \, |2\rangle\langle 3| + \text{h.c.}, \tag{2}$$

and the interaction with the real electric field $\mathcal{E}(z,t)$ is

$$\hat{H}_{\text{int}} = -\hat{d}\,\mathcal{E}. \tag{3}$$

The total Hamiltonian is $\hat{H} = \hat{H}_0 + \hat{H}_{\text{int}}$.

2 Density-Matrix Equations of Motion

The medium is described by the density matrix ρ_{ij}, normalized so that the populations $n_i = \rho_{ii}$ are fractions of the total number density n_0 of the three-level molecules: the number density in level i is $n_0 n_i$, and initially $n_1^0 + n_2^0 + n_3^0 = 1$. The Liouville–von Neumann equation $i\hbar\dot{\hat{\rho}} = [\hat{H}, \hat{\rho}]$ gives, with $\rho_{ij} = \rho_{ji}^*$,

$$\dot{\rho}_{11} = -\frac{i}{\hbar}\,\mathcal{E}\,(\rho_{13}d_{31} - \rho_{31}d_{13}), \tag{4}$$

$$\dot{\rho}_{22} = -\frac{i}{\hbar}\,\mathcal{E}\,(\rho_{23}d_{32} - \rho_{32}d_{23}), \tag{5}$$

$$\dot{\rho}_{33} = \frac{i}{\hbar}\,\mathcal{E}\,(\rho_{13}d_{31} - \rho_{31}d_{13}) + \frac{i}{\hbar}\,\mathcal{E}\,(\rho_{23}d_{32} - \rho_{32}d_{23}), \tag{6}$$

$$\dot{\rho}_{21} = -i\omega_2\,\rho_{21} - \frac{i}{\hbar}\,\mathcal{E}\,(\rho_{23}d_{31} - \rho_{31}d_{23}), \tag{7}$$

$$\dot{\rho}_{31} = -i\omega_3\,\rho_{31} - \frac{i}{\hbar}\,\mathcal{E}\,d_{31}(\rho_{33} - \rho_{11}) + \frac{i}{\hbar}\,\mathcal{E}\,d_{32}\,\rho_{21}, \tag{8}$$

$$\dot{\rho}_{32} = -i(\omega_3 - \omega_2)\,\rho_{32} - \frac{i}{\hbar}\,\mathcal{E}\,d_{32}(\rho_{33} - \rho_{22}) + \frac{i}{\hbar}\,\mathcal{E}\,d_{31}\,\rho_{12}. \tag{9}$$

The coherence ρ_{21} of the forbidden transition is not driven by the field directly, but it is generated through the two allowed transitions.

3 Maxwell Wave Equation

For a field propagating along z and polarized perpendicular to it, with the polarization parallel to the field and no free charges, currents or magnetization, Maxwell's equations give the one-dimensional wave equation

$$\frac{\partial^2 \mathcal{E}}{\partial z^2} - \frac{1}{c^2}\frac{\partial^2 \mathcal{E}}{\partial t^2} = \frac{1}{\epsilon_0 c^2}\frac{\partial^2 \mathcal{P}}{\partial t^2}, \tag{10}$$

with the macroscopic polarization

$$\mathcal{P} = n_0 \operatorname{Tr}(\hat{d}\hat{\rho}) = n_0\big(\rho_{31} d_{13} + \rho_{32} d_{23} + \text{c.c.}\big). \tag{11}$$

Only the coherences of the allowed transitions contribute.

4 Slowly Varying Envelope and Rotating-Wave Approximations

4.1 Envelope decomposition

The field is the sum of two carriers, one resonant with each allowed transition, with slowly varying complex envelopes E_{31} and E_{32}:

$$\mathcal{E}(z,t) = \tfrac{1}{2}\big(E_{31}\, e^{-i\varphi_{31}} + E_{32}\, e^{-i\varphi_{32}}\big) + \text{c.c.}, \qquad \varphi_{3j} = \omega_{3j} t - k_{3j} z, \qquad k_{3j} = \frac{\omega_{3j}}{c}. \tag{12}$$

The coherences are written in the same way,

$$\rho_{31} = \tfrac{1}{2} R_{31}\, e^{-i\varphi_{31}}, \qquad \rho_{32} = \tfrac{1}{2} R_{32}\, e^{-i\varphi_{32}}, \qquad \rho_{21} = \tfrac{1}{2} R_{21}\, e^{-i(\varphi_{31}-\varphi_{32})}, \tag{13}$$

and all envelopes vary slowly compared with their carriers (SVEA):

$$\left|\frac{\partial E_{3j}}{\partial t}\right| \ll \omega_{3j}|E_{3j}|, \qquad \left|\frac{\partial E_{3j}}{\partial z}\right| \ll k_{3j}|E_{3j}|, \tag{14}$$

and likewise for R_{3j} [2, 3]. In the rotating-wave approximation (RWA) the terms oscillating at sums of carrier frequencies, and the terms in which a field drives the transition it is not resonant with, are dropped. The detunings

$$\delta\omega_{31} = \omega_{31} - \omega_3, \qquad \delta\omega_{32} = \omega_{32} - (\omega_3 - \omega_2), \qquad \delta\omega_{21} = \omega_{31} - \omega_{32} - \omega_2 \tag{15}$$

appear as $i\,\delta\omega_{ij} R_{ij}$ terms; we take exact resonance, $\delta\omega_{ij} = 0$.

4.2 Bloch equations for the envelopes

Substituting the envelopes into the density-matrix equations and keeping the resonant terms gives

$$\dot{n}_1 = \tfrac{1}{2}\,\mathrm{Re}\left[\left(-\tfrac{id_{31}}{\hbar}E_{31}\right)^* R_{31}\right], \tag{16}$$

$$\dot{n}_2 = \tfrac{1}{2}\,\mathrm{Re}\left[\left(-\tfrac{id_{32}}{\hbar}E_{32}\right)^* R_{32}\right], \tag{17}$$

$$\dot{n}_3 = -\tfrac{1}{2}\,\mathrm{Re}\left[\left(-\tfrac{id_{31}}{\hbar}E_{31}\right)^* R_{31} + \left(-\tfrac{id_{32}}{\hbar}E_{32}\right)^* R_{32}\right], \tag{18}$$

$$\dot{R}_{21} = \tfrac{1}{2}\left[\left(-\tfrac{id_{31}}{\hbar}E_{31}\right) R_{32}^* + \left(-\tfrac{id_{32}}{\hbar}E_{32}\right)^* R_{31}\right], \tag{19}$$

$$\dot{R}_{31} = \left(-\tfrac{id_{31}}{\hbar}E_{31}\right)(n_3 - n_1) - \tfrac{1}{2}\left(-\tfrac{id_{32}}{\hbar}E_{32}\right) R_{21}, \tag{20}$$

$$\dot{R}_{32} = \left(-\tfrac{id_{32}}{\hbar}E_{32}\right)(n_3 - n_2) - \tfrac{1}{2}\left(-\tfrac{id_{31}}{\hbar}E_{31}\right) R_{21}^*. \tag{21}$$

The combinations $-id_{3j}E_{3j}/\hbar$ (Rabi frequencies) appear throughout; they define the dimensionless fields in Section 5. Populations are conserved, $\dot{n}_1 + \dot{n}_2 + \dot{n}_3 = 0$.

4.3 Field equations

Substituting the field and the polarization into the wave equation (10), the terms at each carrier balance separately. Dropping second derivatives of the envelopes (SVEA),

$$\frac{\partial E_{3j}}{\partial z} + \frac{1}{c}\frac{\partial E_{3j}}{\partial t} = i\,\frac{n_0 k_{3j} d_{j3}}{2\epsilon_0}\,R_{3j}, \qquad j = 1, 2. \tag{22}$$

5 Retarded Time and Dimensionless Form

5.1 Retarded time

In the frame moving with the fields, $z' = z$ and $t_{\mathrm{ret}} = t - z/c$, the operator $\partial_z + c^{-1}\partial_t$ becomes $\partial_{z'}$ and the field equations are pure propagation equations,

$$\frac{\partial E_{3j}}{\partial z'} = i\,\frac{n_0 k_{3j} d_{j3}}{2\epsilon_0}\,R_{3j}, \tag{23}$$

while the Bloch equations keep their form with $t \to t_{\mathrm{ret}}$.

5.2 Scaled variables

Lengths are scaled by the sample length L and times by T_0,

$$\xi = \frac{z'}{L}, \qquad \tau = \frac{t_{\mathrm{ret}}}{T_0}, \qquad T_0 = \frac{2\epsilon_0\hbar}{n_0 L k_{32}|d_{32}|^2}, \tag{24}$$

and the fields by their Rabi frequencies,

$$\boxed{\tilde{E}_{3j} = -i\,\frac{T_0 d_{3j}}{\hbar}\,E_{3j}.} \tag{25}$$

The field equations become

$$\frac{\partial \tilde{E}_{31}}{\partial \xi} = \frac{n_0 L |d_{31}|^2 k_{31} T_0}{2\epsilon_0 \hbar} R_{31} = \eta\, R_{31}, \qquad \frac{\partial \tilde{E}_{32}}{\partial \xi} = \frac{n_0 L |d_{32}|^2 k_{32} T_0}{2\epsilon_0 \hbar} R_{32} = R_{32}, \tag{26}$$

where T_0 was chosen to make the coupling of the $3 \to 2$ field one, and the coupling of the $3 \to 1$ field is

$$\boxed{\eta = \left|\frac{d_{31}}{d_{32}}\right|^2 \frac{k_{31}}{k_{32}} = d^2\,\omega, \qquad d = \left|\frac{d_{31}}{d_{32}}\right|, \qquad \omega = \frac{k_{31}}{k_{32}}.} \tag{27}$$

η is fixed by the spectroscopy of the two transitions; it is not a free parameter.

Note

Sign of the field equation. Here $\partial\tilde{E}/\partial\xi = +\eta R$, while the two-level model has $\partial A'/\partial z' = -\eta R$. Both use the same carrier $e^{i(kz-\omega t)}$ and the same coherence envelope (upper–lower element $= \frac{1}{2} R\, e^{i(kz-\omega t)}$); they differ only in the phase of the field scale. The two-level model uses $A' = A/A_0$ with $A_0 = -i\hbar/(d\,T_0)$, i.e. $A' = +i\,d\,T_0 A/\hbar$, while Eq. (25) has $-i$. For the same physical field $A' = -\tilde{E}$, which flips the sign of every field term: the two-level equations $\dot{w} = A'R$, $\dot{R} = -A'w$, $\partial_{z'} A' = -\eta R$ correspond here to $\dot{n}_{32} = -\tilde{E}_{32} R_{32}$, $\dot{R}_{32} = \tilde{E}_{32} n_{32}$, $\partial_\xi \tilde{E}_{32} = R_{32}$. Intensities $|\tilde{E}|^2$ are the same in both conventions.

5.3 Relaxation and pumping

Non-coherent processes are added phenomenologically. The population differences

$$n_{31} = n_3 - n_1, \qquad n_{32} = n_3 - n_2 \tag{28}$$

relax towards their reference values n^0_{3j} with times $T^{(1)}_{3j}$, and the coherences R_{ij} dephase towards R^0_{ij} with times $T^{(2)}_{ij}$, $ij = 31, 32, 21$. Positive n_{3j} means inversion on the $3 \to j$ transition. Pumps Λ_{31} and Λ_{32} act on the two population differences. In scaled form,

$$\tau^{(k)}_{ij} = \frac{T^{(k)}_{ij}}{T_0}, \qquad \tilde{\Lambda}_{3j} = T_0\, \Lambda_{3j}. \tag{29}$$

5.4 Final dimensionless equations

Combining the Bloch equations for $n_3 - n_1$ and $n_3 - n_2$ and using Eq. (25):

$$\frac{\partial n_{31}}{\partial \tau} = -\operatorname{Re}\left(\tilde{E}_{31}^{*} R_{31} + \tfrac{1}{2}\tilde{E}_{32}^{*} R_{32}\right) - \frac{n_{31} - n_{31}^{0}}{\tau_{31}^{(1)}} + \tilde{\Lambda}_{31}, \tag{30}$$

$$\frac{\partial n_{32}}{\partial \tau} = -\operatorname{Re}\left(\tfrac{1}{2}\tilde{E}_{31}^{*} R_{31} + \tilde{E}_{32}^{*} R_{32}\right) - \frac{n_{32} - n_{32}^{0}}{\tau_{32}^{(1)}} + \tilde{\Lambda}_{32}, \tag{31}$$

$$\frac{\partial R_{31}}{\partial \tau} = \tilde{E}_{31}\, n_{31} - \tfrac{1}{2}\tilde{E}_{32}\, R_{21} - \frac{R_{31} - R_{31}^{0}}{\tau_{31}^{(2)}}, \tag{32}$$

$$\frac{\partial R_{32}}{\partial \tau} = \tilde{E}_{32}\, n_{32} - \tfrac{1}{2}\tilde{E}_{31}\, R_{21}^{*} - \frac{R_{32} - R_{32}^{0}}{\tau_{32}^{(2)}}, \tag{33}$$

$$\frac{\partial R_{21}}{\partial \tau} = \tfrac{1}{2}\left(\tilde{E}_{31} R_{32}^{*} + \tilde{E}_{32}^{*} R_{31}\right) - \frac{R_{21} - R_{21}^{0}}{\tau_{21}^{(2)}}, \tag{34}$$

$$\frac{\partial \tilde{E}_{31}}{\partial \xi} = \eta\, R_{31}, \qquad \frac{\partial \tilde{E}_{32}}{\partial \xi} = R_{32}. \tag{35}$$

In the paper the fields are written $\tilde{E}_1 \equiv \tilde{E}_{31}$ and $\tilde{E}_2 \equiv \tilde{E}_{32}$, and the pumps are absorbed into the reference values n_{3j}^{0} except for the flare pump of Section 9.

5.5 Real-variable reduction

If the initial coherences R_{ij}^{0} and the entrance fields $\tilde{E}_{3j}(0,\tau)$ are real, the right-hand sides of Eqs. (30)–(35) are real whenever the variables are, so all variables stay real. Then $\operatorname{Re}(\tilde{E}^{*}R) = \tilde{E}R$, $R_{21}^{*} = R_{21}$, and the system becomes

$$\frac{\partial n_{31}}{\partial \tau} = -\left(\tilde{E}_{31} R_{31} + \tfrac{1}{2}\tilde{E}_{32} R_{32}\right) - \frac{n_{31} - n_{31}^{0}}{\tau_{31}^{(1)}} + \tilde{\Lambda}_{31}, \tag{36}$$

$$\frac{\partial n_{32}}{\partial \tau} = -\left(\tfrac{1}{2}\tilde{E}_{31} R_{31} + \tilde{E}_{32} R_{32}\right) - \frac{n_{32} - n_{32}^{0}}{\tau_{32}^{(1)}} + \tilde{\Lambda}_{32}, \tag{37}$$

$$\frac{\partial R_{31}}{\partial \tau} = \tilde{E}_{31}\, n_{31} - \tfrac{1}{2}\tilde{E}_{32}\, R_{21} - \frac{R_{31} - R_{31}^{0}}{\tau_{31}^{(2)}}, \tag{38}$$

$$\frac{\partial R_{32}}{\partial \tau} = \tilde{E}_{32}\, n_{32} - \tfrac{1}{2}\tilde{E}_{31}\, R_{21} - \frac{R_{32} - R_{32}^{0}}{\tau_{32}^{(2)}}, \tag{39}$$

$$\frac{\partial R_{21}}{\partial \tau} = \tfrac{1}{2}\left(\tilde{E}_{31} R_{32} + \tilde{E}_{32} R_{31}\right) - \frac{R_{21} - R_{21}^{0}}{\tau_{21}^{(2)}}, \tag{40}$$

$$\frac{\partial \tilde{E}_{31}}{\partial \xi} = \eta\, R_{31}, \qquad \frac{\partial \tilde{E}_{32}}{\partial \xi} = R_{32}. \tag{41}$$

Note

Equations (36)–(41) are the system solved by `mbsolver.three_level.solve_1d`: fourth-order Runge–Kutta in τ at each ξ, and an explicit step in ξ for the fields. They hold unchanged when the relaxation times depend on time.

6 Initial and Boundary Conditions

6.1 Populations

The initial populations are given by the ratios

$$r_1 = \frac{n_3^0}{n_1^0}, \qquad r_2 = \frac{n_3^0}{n_2^0}, \tag{42}$$

which, with $n_1^0 + n_2^0 + n_3^0 = 1$, give

$$n_1^0 = \frac{r_2}{r_1 + r_2 + r_1 r_2}, \qquad n_2^0 = \frac{r_1}{r_1 + r_2 + r_1 r_2}, \qquad n_3^0 = \frac{r_1 r_2}{r_1 + r_2 + r_1 r_2}, \tag{43}$$

and the population differences

$$n_{31}^0 = \frac{r_2(r_1 - 1)}{r_1 + r_2 + r_1 r_2}, \qquad n_{32}^0 = \frac{r_1(r_2 - 1)}{r_1 + r_2 + r_1 r_2}. \tag{44}$$

The $3 \to j$ transition is inverted when $r_j > 1$. The mixed configuration of the paper, an inverted $3 \to 2$ transition with a non-inverted $3 \to 1$ leakage pathway, is $r_1 < 1 < r_2$. The reference values n_{3j}^0 of the relaxation terms are these initial values, the same at every ξ. All physical initial states lie in the triangle of Fig. 2.

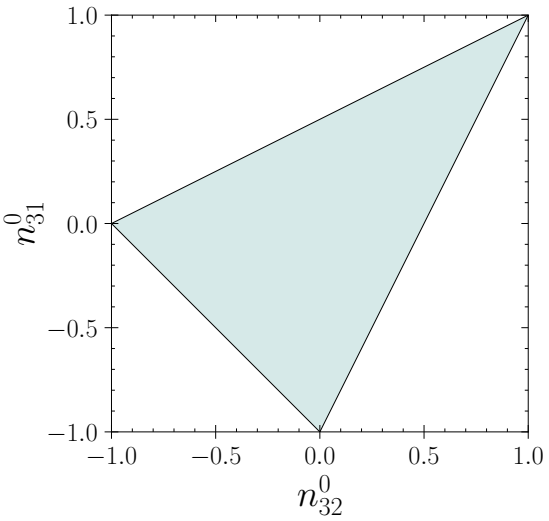


Figure 2: Allowed initial population differences. The corners are all molecules in $|3\rangle$ $(1, 1)$, in $|2\rangle$ $(-1, 0)$ and in $|1\rangle$ $(0, -1)$.

6.2 Coherences

The magnitude of each initial coherence is bounded by the populations (Cauchy–Schwarz),

$$|R_{ij}^0| \leq \mathcal{R}_{ij}^{\max} = 2\sqrt{n_i^0 n_j^0}, \tag{45}$$

and is given as a fraction c_{ij} of this bound,

$$R_{ij}^0 = c_{ij}\,\mathcal{R}_{ij}^{\max}, \qquad |c_{ij}| \le 1. \tag{46}$$

Since no field drives the forbidden transition, the paper takes $c_{21} = 0$. In terms of r_1 and r_2,

$$\mathcal{R}_{31}^{\max} = \frac{2r_2\sqrt{r_1}}{r_1 + r_2 + r_1 r_2}, \qquad \mathcal{R}_{32}^{\max} = \frac{2r_1\sqrt{r_2}}{r_1 + r_2 + r_1 r_2}, \qquad \mathcal{R}_{21}^{\max} = \frac{2\sqrt{r_1 r_2}}{r_1 + r_2 + r_1 r_2}. \tag{47}$$

6.3 Boundary conditions

The fields entering the sample, $\tilde{E}_{3j}(0,\tau)$, are given; for emission generated by the medium itself they are zero. With uniform initial coherences, Eq. (41) gives the fields at the exit at $\tau = 0$: $\tilde{E}_{31}(1,0) = \tilde{E}_{31}(0,0) + \eta R_{31}^0$ and $\tilde{E}_{32}(1,0) = \tilde{E}_{32}(0,0) + R_{32}^0$.

7 Physical Scales

7.1 Time scale

With the Einstein coefficient of the $3 \to 2$ transition,

$$\Gamma_{32} = \frac{\omega_{32}^3 |d_{32}|^2}{3\pi\epsilon_0\hbar c^3} = \frac{k_{32}^3 |d_{32}|^2}{3\pi\epsilon_0\hbar}, \tag{48}$$

the time scale (24) can be written in the two-level form

$$T_0 = \frac{2\epsilon_0\hbar}{n_0 L k_{32} |d_{32}|^2} = \frac{2k_{32}^2}{3\pi\,\Gamma_{32}\,n_0 L} = \frac{8\pi}{3\,\Gamma_{32}\,n_0 L \lambda_{32}^2}. \tag{49}$$

Both forms agree when Γ_{32} is computed from d_{32}; with a measured decay rate they differ by the ratio of the two rates.

7.2 Fields, intensities and pump

The dimensionless fields are Rabi frequencies, Eq. (25). In units of the field scale of the $3 \to 2$ transition,

$$E_0 = \frac{\hbar}{T_0 |d_{32}|}, \tag{50}$$

the field amplitudes are $|E_{32}|/E_0 = |\tilde{E}_{32}|$ and $|E_{31}|/E_0 = |\tilde{E}_{31}|/d$. Both intensities are therefore given in the same unit,

$$I_{31} = I_0 \frac{|\tilde{E}_{31}|^2}{d^2}, \qquad I_{32} = I_0\,|\tilde{E}_{32}|^2, \qquad I_0 = \tfrac{1}{2}\epsilon_0 c E_0^2, \tag{51}$$

and a pump rate Λ in $\mathrm{m^{-3}\,s^{-1}}$ (molecules per volume and time) corresponds to the dimensionless pump $\tilde{\Lambda} = \Lambda/\Lambda_0$ with $\Lambda_0 = n_0/T_0$.

8 In the Code

8.1 Solver

`mbsolver.three_level.solve_1d` takes the dimensionless parameters (times in units of T_0) and returns the variables on `n_planes` evenly spaced planes $\xi \in [0, 1]$:

Symbol	Argument	Result
τ grid, ξ grid	`t_max`, `nt`, `nz`	`res.time`, `res.z`
d, ω ($\eta = \omega d^2$)	`d`, `omega`	—
r_1, r_2	`r1`, `r2`	—
c_{31}, c_{32}, c_{21}	`c31`, `c32`, `c21` (default 0)	—
$\tau_{31}^{(1)}$, $\tau_{32}^{(1)}$	`t1_31`, `t1_32`	`res.t1_31`, `res.t1_32`
$\tau_{31}^{(2)}$, $\tau_{32}^{(2)}$, $\tau_{21}^{(2)}$	`t2_31`, `t2_32`, `t2_21` (default ∞)	`res.t2_31`, …
$\tilde{\Lambda}_{31}$, $\tilde{\Lambda}_{32}$	`pump31`, `pump32`	`res.pump31`, `res.pump32`
$\tilde{E}_{31}(0,\tau)$, $\tilde{E}_{32}(0,\tau)$	`field31_in`, `field32_in`	`res.field31_in`, …
$\tilde{E}_{31}$, $\tilde{E}_{32}$	—	`res.e31`, `res.e32`
$\lvert\tilde{E}_{31}\rvert^2/d^2$, $\lvert\tilde{E}_{32}\rvert^2$	—	`res.intensity31`, `res.intensity32`
n_{31}, n_{32}	—	`res.n31`, `res.n32`
R_{31}, R_{32}, R_{21}	—	`res.r31`, `res.r32`, `res.r21`

Relaxation times are numbers or `Steps` (`math.inf` for none); pumps and entrance fields are `Pulses`, numbers or lists of them.

8.2 Helpers

`mbsolver.three_level.populations(r1, r2)` returns n_1^0, n_2^0, n_3^0, n_{31}^0, n_{32}^0 and $\mathcal{R}_{31}^{\max}$, $\mathcal{R}_{32}^{\max}$, $\mathcal{R}_{21}^{\max}$ (Section 6).

`mbsolver.three_level.scaling(nu31, nu32, d31, d32, l, n0, gamma=None)` takes the frequencies $\nu_{3j} = \omega_{3j}/2\pi$ (Hz), the dipole moments (C m; `mbsolver.constants.DEBYE` converts from debye), L (m) and n_0 (m^{-3}) and returns d, ω, η, k_{3j}, λ_{3j}, Γ_{32}, T_0, E_0, I_0, $10^{26} I_0$ (`i0_jy`, used to plot intensities in Jy) and Λ_0 (Section 7). By default Γ_{32} is computed from d_{32}; `gamma` replaces it with a given rate.

9 Example: S255IR-NIRS3

The 6.7 GHz methanol flare in S255IR-NIRS3 was observed with the Toruń 32 m telescope [8] during the accretion burst of the central massive protostar [7]. Rajabi et al. [9] modelled it as Dicke superradiance [1] with the two-level Maxwell–Bloch equations used for astrophysical masers [4, 5, 6]. The three-level model adds the 239.7 GHz transition, which shares the upper level of the 6.7 GHz line and decays more than four orders of magnitude faster.

Transition	Levels	ν (MHz)	d (D)
$3 \to 1$	$5_1\,A^+ \to 4_1\,A^+$	239 746.219	0.59425
$3 \to 2$	$5_1\,A^+ \to 6_0\,A^+$	6 668.564	0.67936

These give $d = 0.8747$, $\omega = 35.95$ and $\eta = 27.51$. From d_{3j}, $\Gamma_{31} = 5.66 \times 10^{-5}$ s^{-1} and $\Gamma_{32} = 1.59 \times 10^{-9}$ s^{-1}.

The sample values are those of the two-level fit: $L = 2.1 \times 10^{13}$ m, $\Gamma_{32} = 1.56 \times 10^{-9}$ s^{-1}, initial Bloch angle $\theta_0 = 4.14 \times 10^{-10}$ and density $n_0^{(2)} = 1.17 \times 10^{-6}$ m^{-3} of molecules in levels 2 and 3. The two-level inversion $\cos\theta_0 = (n_3 - n_2)/(n_3 + n_2)$ fixes

$$r_2 = \frac{1 + \cos\theta_0}{1 - \cos\theta_0} = \cot^2\frac{\theta_0}{2} \simeq 2.33 \times 10^{19}, \tag{52}$$

and the three-level density counts level 1 as well,

$$n_0 = n_0^{(2)}\,\frac{n_1 + n_2 + n_3}{n_2 + n_3} = n_0^{(2)}\,\frac{r_1 + r_2 + r_1 r_2}{r_1(1 + r_2)}. \tag{53}$$

With $r_1 = 0.94$, $c_{31} = 10^{-10}$ and $c_{32} = 1$, this gives $n_0 = 2.41 \times 10^{-6}$ m^{-3} and, with the measured Γ_{32}, $T_0 \simeq 5.24 \times 10^4$ s. The relaxation times $T_{32}^{(1)} = 1.64 \times 10^7$ s and $T_{32}^{(2)} = 1.55 \times 10^6$ s are those of the two-level fit, and the $3 \to 1$ times are ten times shorter. The flare is driven by the pump $\Lambda_{32}(t) = \Lambda_1\,\mathrm{sech}^2[(t - t_p)/T_p]$ with $\Lambda_1 = 2.6 \times 10^{-13}$ m^{-3} s^{-1} and $T_p = 8.1 \times 10^6$ s ($\tau_p = T_p/T_0 \simeq 155$), centred at $t_p = 795.5\,T_0$.

`examples/three_level/s255_flare.py` runs this case for $r_1 = 0.79$, 0.89 and 0.94 and plots both intensities in Jy with the observed 6.7 GHz flux of [8].

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